SpanLink Final Presentation

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SpanLink

01

Design Document: High-Level Overview

Provider Workflow

A provider/administrator of the test will be able to login or sign up from our homepage and then be routed to their dashboard. A link generator on the provider's dashboard will allow a provider to create a unique, one-time use link that generates a test for a patient. The provider can then send that link to a patient with the test in their preferred language with appropriate color blind settings. In addition to creating links for patients, providers will also be able to view any ongoing tests or completed tests that they have created, see notifications when a test is completed, export individual test data, and view aggregated statistics for all tests generated and individual metrics via a GUI and graph features.

Patient Experience

A patient will not be able to login or sign up to our application, as the patient will only be utilizing our application for diagnostic purposes. The administering provider will analyze and discuss results with the patient outside of our platform for now. When a user clicks their provided link, they will be directed to a page where patients will be able to adjust any accessibility settings that would aid them with any disabilities, handicaps, or preferences that they may have, such as: color blindness, vision impairments, or language preference. Once they click Next, the web page will deliver instructions as well as provide a practice test for the patient before they start the real test. Once the user has completed the test, they will be presented with the option to take a feedback survey to help us better serve our customers.

How was SpanLink Built?

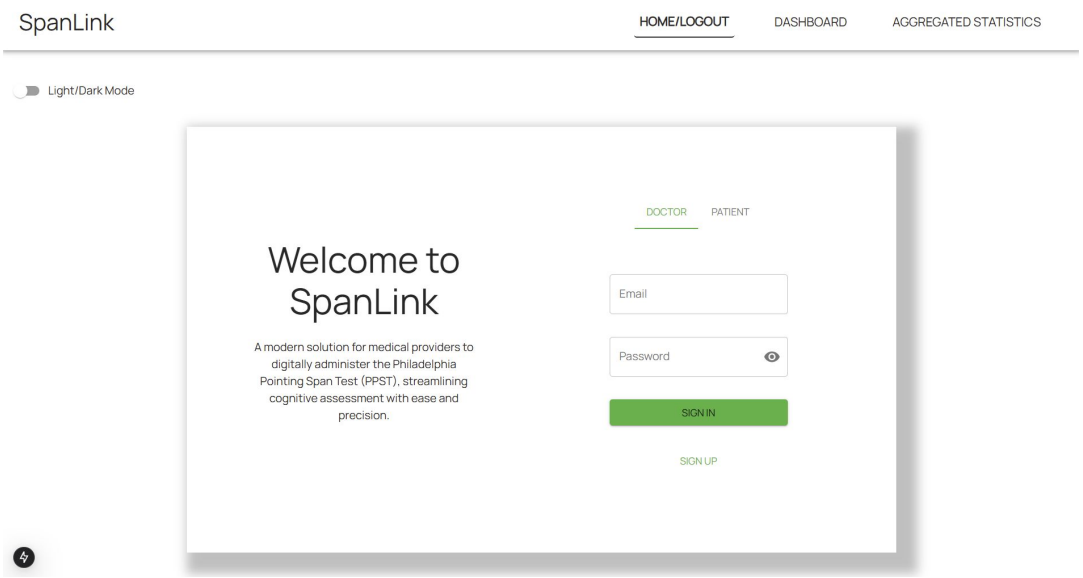
- Frontend
  - React & Material UI: Frontend UI components
  - Howler.js: For audio feedback
  - File-Saver: Saving files
- Backend
  - Django: Backend Framework
  - Openpyxl: Reading/writing to Excel Files
  - Seaborn, Numpy, & Scipy: Statistic graphs

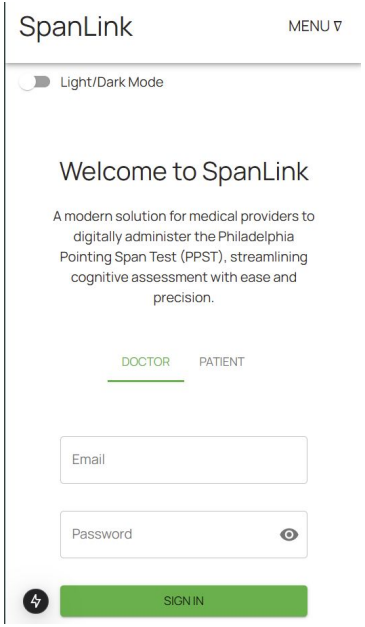
02

Project Specification

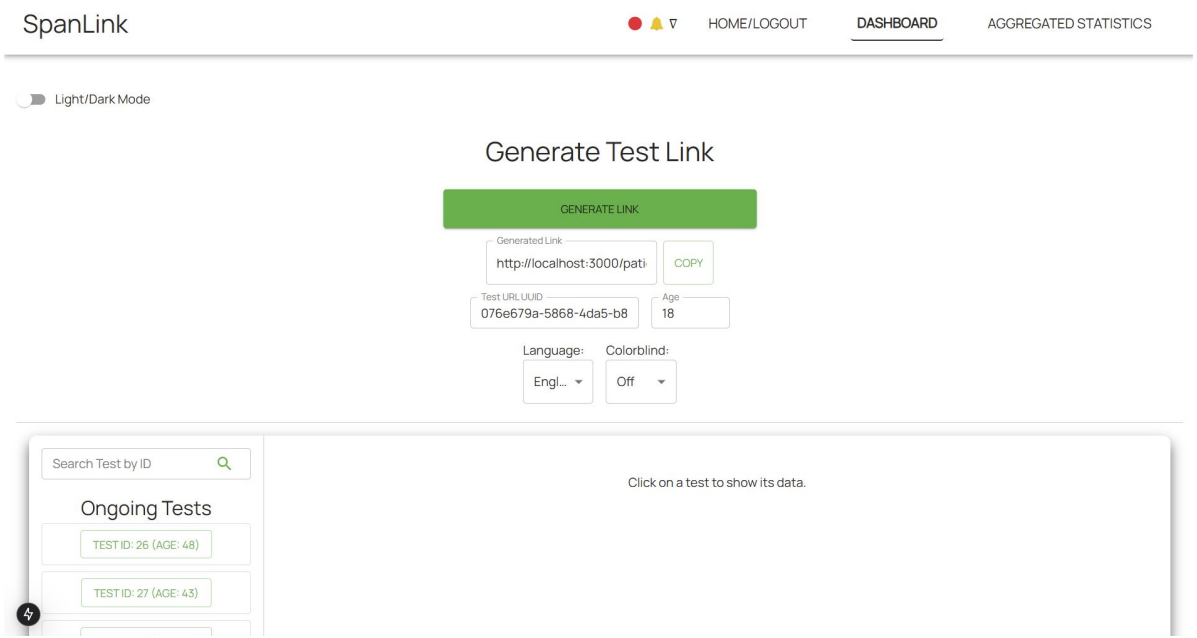
Web Based

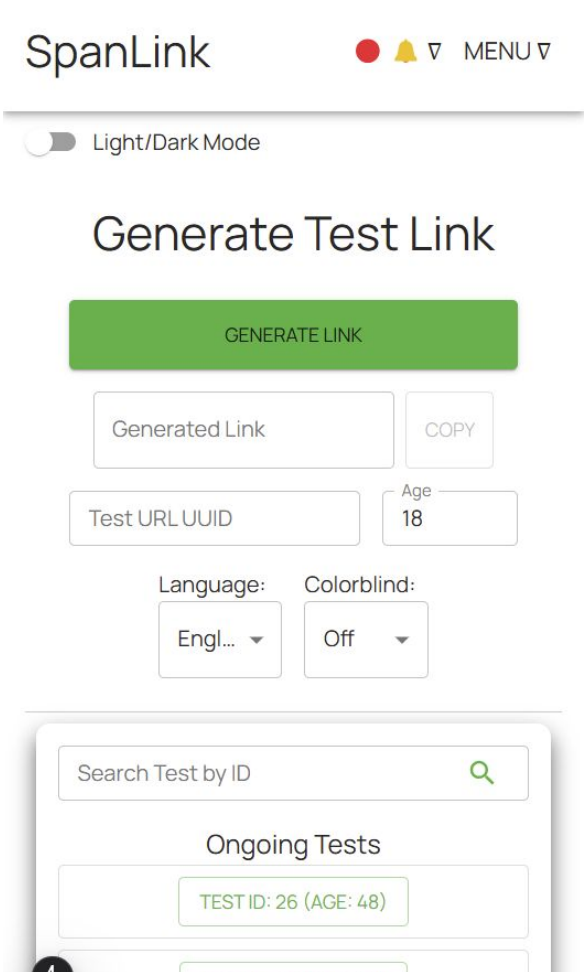
1. Our service will be available as a web application, optimized to be responsive for all screen sizes and at least one browser from each major OS (Windows, MacOS, iOS, and Android).



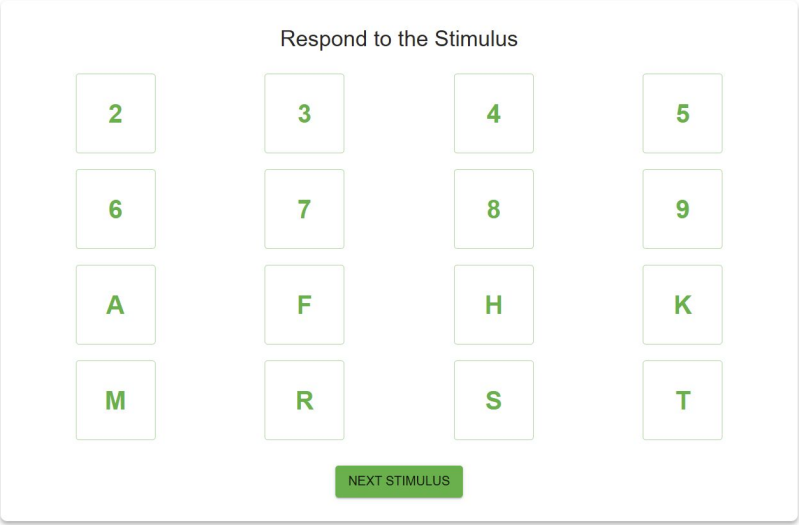


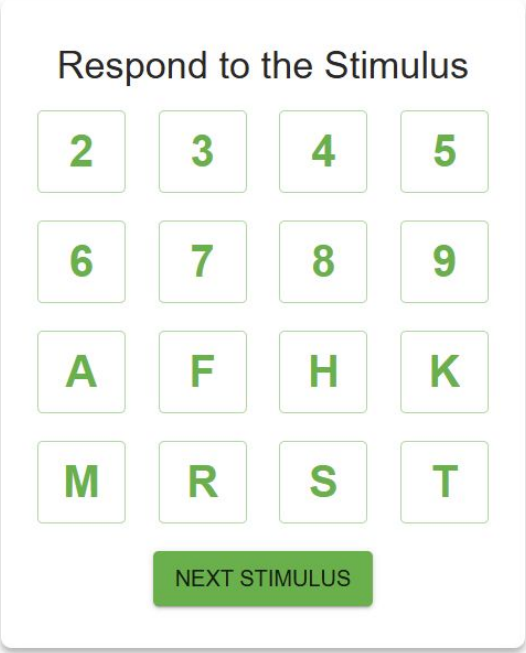
Web Based cont.





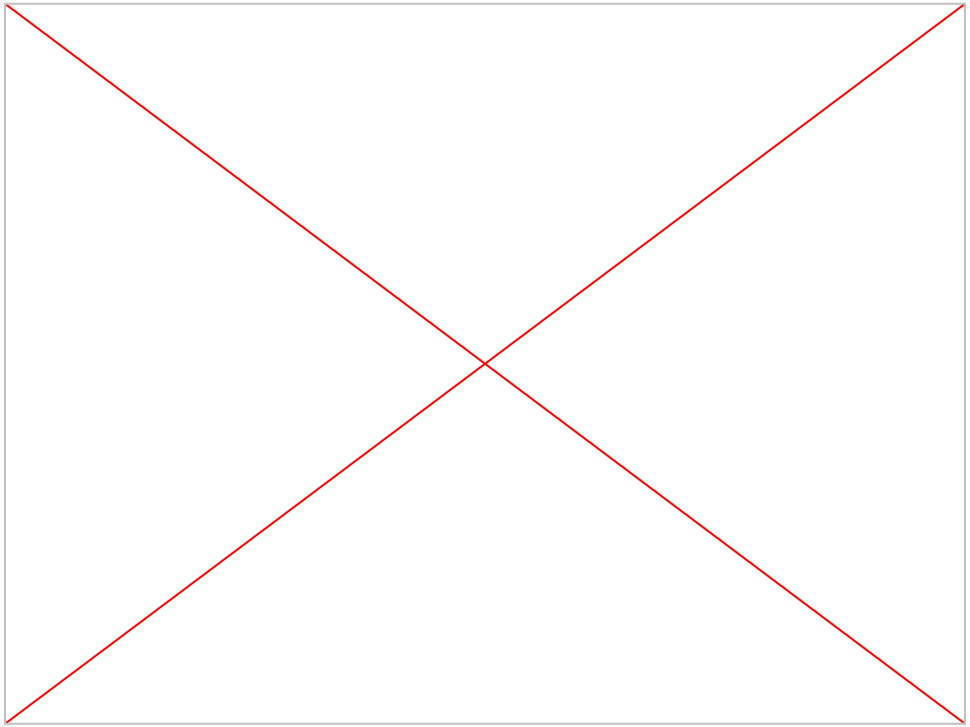
Web Based cont.





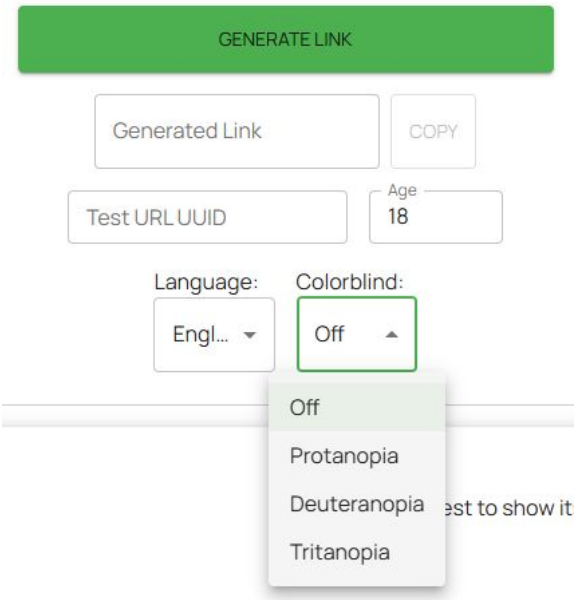
Login

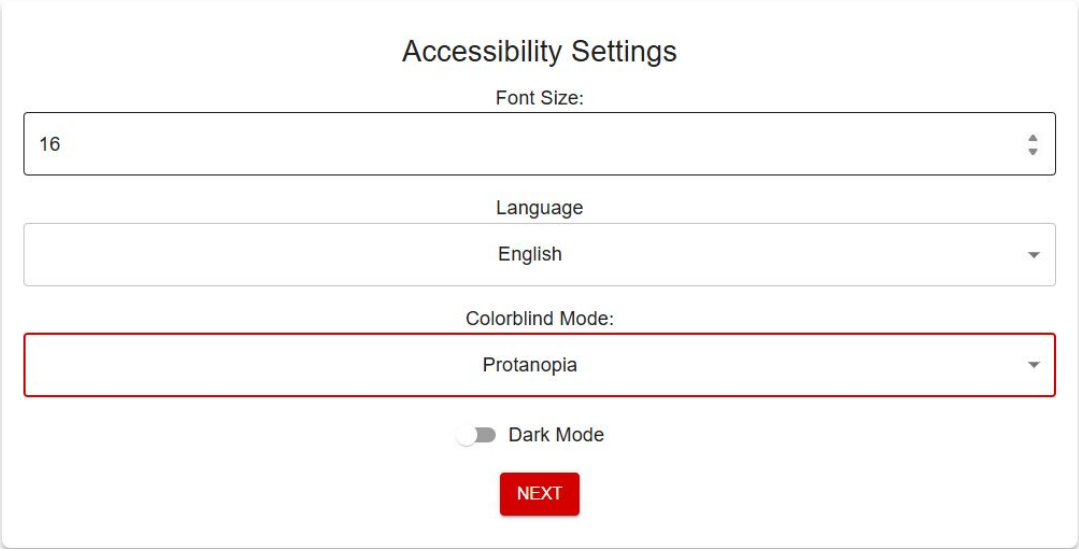
2. Login: Test givers will be able to create an account and login to the SpanLink portal using an email/username and password login system. Web administrators will also be able to log in to the system to view statistics and users. Test takers will not have accounts.



Accessibility

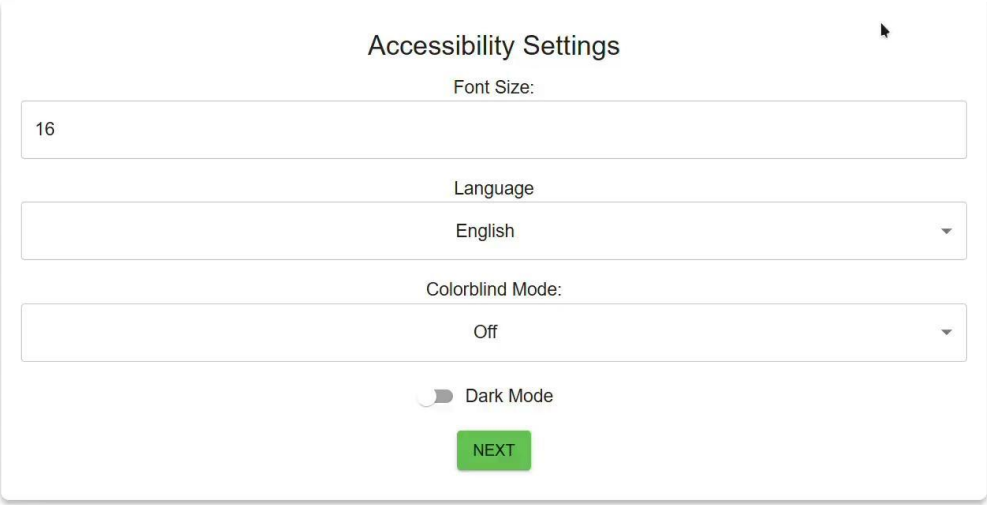
3. The stimuli will be presented to the user audibly and visually to accommodate vision or hearing impairment(s). Audio and visual effects will also be used to help test takers know when a symbol has been successfully pressed. The test will also be available in two languages, English and Spanish. Language and colorblind modes can be set for patient tests by their providers, and also altered within the test if the settings are not to the user's liking.





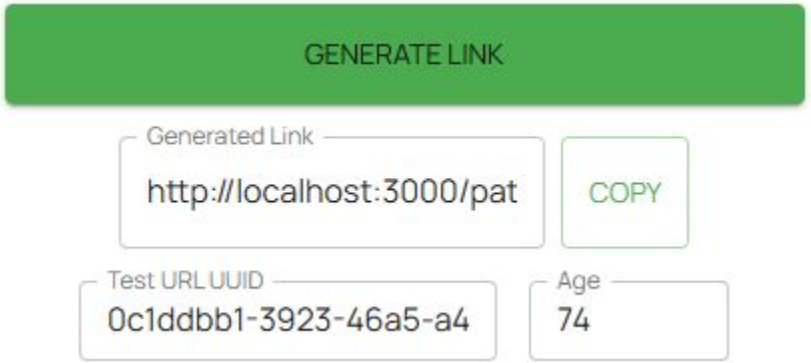
Customization

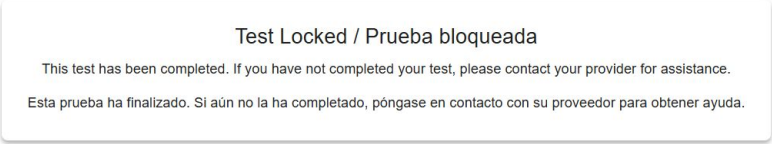
7. Before a user takes a test, they will be able to choose between an English or Spanish voice and change the theme from light and dark mode (for accessibility)



Web Links

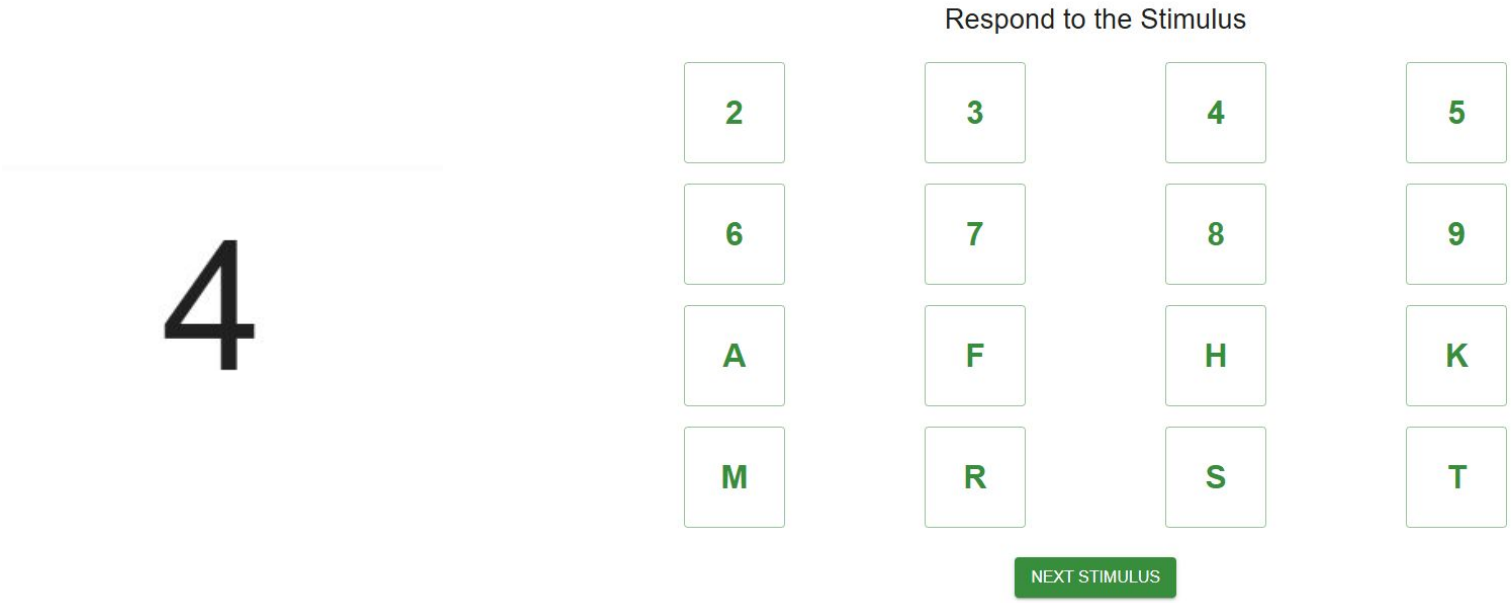
4. The test giver will be able to generate and share a link with a user via their email. A test receiver will be able to follow this link to access and complete a PPST. Only a recipient of a link will be able to access a test. Once completed, tests will become inaccessible from the same url in order to preserve patient data.





Test

5. Each instance of the Philadelphia Pointing Span Test (PPST) will comprise of 12 stimuli in total: 6 with digits and 6 with a mix of digits and letters. For each group of 6 stimuli, 3 will be 4-span and 3 will be 5-span, where span is the number of symbols given to the user for recollection. Before each set, 2 practice stimuli will be administered. The goal is for the test taker to press the correct symbols in ascending order. The stimuli for each test will be the same for each test taker, but order of each set may vary.



Test Results

6. Once a test taker completes the PPST, the test giver will be able to view and download the test results in spreadsheet format. Only the test giver will have access to the results. Data such as which symbols were pressed in which order, response latency, overall test duration, and correct response percentage will be recorded (practice stimuli not included in results).

Individual Statistics

	A	B	C	D	E	F
1	Response #	Stimulus Sequence	Response Sequence	Individual Latencies (ms)	Average Latency (ms)	Correct
2	1	4, 2, 9, 6	2, 4, 6, 9	3737, 800, 807, 718	1515.5	Yes
3	2	4, 2, 9, 6, 7	4, 2, 6, 7, 9	1625, 818, 651, 544, 675	862.6	No
4	3	2, 5, 3, 7	2, 3, 5, 7	1454, 643, 721, 690	877	Yes
5	4	3, 6, 2, 8	2, 3, 6, 8	1551, 697, 750, 807	951.25	Yes
6	5	6, 2, 4, 8	2, 4, 6, 8	1323, 965, 822, 750	965	Yes
7	6	3, 7, 5, 9, 8	5, 7, 8, 9	1899, 1145, 655, 617	1079	No
8	7	5, 3, 8, 2, 9	2, 3, 5, 8, 9	1011, 644, 895, 1065, 587	840.4	Yes
9	8	4, 7, 9, 3, 6	3, 4, 6, 7, 9	1753, 680, 897, 1299, 753	1076.4	Yes
10	9	F, 2, A, 7	2, 7, A, F	1159, 759, 664, 567	787.25	Yes
11	10	3, S, 2, A, 7	2, 3, 7, A, S	1050, 589, 670, 1504, 782	919	Yes
12	11	A, 2, R, 9	2, 9, A, R	835, 862, 955, 739	847.75	Yes
13	12	8, H, 3, F	3, 8, F, H	1253, 781, 838, 800	918	Yes
14	13	7, R, T, 4	4, 7, R, T	1818, 793, 1139, 741	1122.75	No
15	14	7, R, 6, T, 4	4, 6, 7, R, T	1277, 768, 707, 818, 790	872	No
16	15	5, S, M, 6, 2	2, 4, 6, M, S	1313, 704, 1025, 1498, 707	1049.4	No
17	16	7, 5, 4, R, K	4, 5, 7, K, R	1445, 691, 941, 1130, 810	1003.4	Yes

Test Results cont.

Aggregated Statistics

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Test ID	Patient Age	Stimuli Order	Correct Sequence	Response Sequence	Sequence Latency	Average Latency	Correct Response	Count Correct	Span Length	Test Created At	Test Started At	Test Completed At	First Latency
2	1	18		['4', '5', '7', 'K', 'R']	['4', '5', '7', 'K', 'R']	[2802, 2585, 2505, 2754, 250	2630.2	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	2802
3	1	18		['4', '7', 'T', 'R']	['4', '7', 'T', 'R']	[1337, 1132, 1470, 1367]	1326.5	TRUE	4	4	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	1337
4	1	18		['4', '5', '7', 'K', 'R']	['4', '5', '7', 'K', 'R']	[3159, 3074, 2869, 3019, 311	3047.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	3159
5	1	18		['2', '3', '6', '8']	['2', '3', '6', '8']	[4271, 4438, 4213, 4149]	4267.75	TRUE	4	4	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	4271
6	1	18		['2', '3', '5', '7']	['2', '3', '5', '7']	[2462, 2569, 2548, 2263]	2460.5	TRUE	4	4	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	2462
7	1	18		['3', '4', '6', '7', '9']	['3', '4', '6', '7', '9']	[3808, 3681, 3572, 3698, 350	3651.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	3808
8	1	18		['3', '8', 'F', 'H']	['3', '8', 'F', 'H']	[1328, 1288, 1234, 1017]	1216.75	TRUE	4	4	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	1328
9	1	18		['4', '7', 'T', 'R']	['4', '7', 'T', 'R']	[1163, 1235, 1101, 1394]	1223.25	TRUE	4	4	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	1163
10	1	18		['2', '5', '6', 'M', 'S'	['2', '5', '6', 'M', 'S']	[1853, 1959, 1855, 1824, 194	1886.6	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	1853
11	1	18		['3', '4', '6', '7', '9']	['3', '4', '6', '7', '9']	[1252, 1088, 1295, 1230, 110	1193.4	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	1252
12	1	18		['3', '5', '7', '8', '9']	['3', '5', '7', '8', '9']	[3408, 3335, 3491, 3307, 314	3336.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	3408
13	1	18		['3', '4', '6', '7', '9']	['3', '4', '6', '7', '9']	[1634, 1715, 1823, 1730, 172	1724.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 10:25:4	29-04-25 10:57:40	1634
14	2	21		['3', '5', '7', '8', '9']	['3', '5', '7', '8', '9']	[3252, 3106, 3067, 3014, 288	3065.2	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	3252
15	2	21		['2', '3', '6', '8']	['2', '3', '6', '8']	[1825, 1706, 1667, 2023]	1805.25	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1825
16	2	21		['2', '3', '5', '7']	['2', '3', '5', '7']	[2732, 2938, 2815, 2934]	2854.75	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	2732
17	2	21		['4', '7', 'T', 'R']	['4', '7', 'T', 'R']	[1498, 1603, 1479, 1739]	1579.75	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1498
18	2	21		['2', '3', '5', '7']	['2', '3', '5', '7']	[1091, 1003, 1378, 1082]	1138.5	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1091
19	2	21		['4', '5', '7', 'K', 'R']	['4', '5', '7', 'K', 'R']	[1486, 1521, 1547, 1614, 145	1523.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1486
20	2	21		['2', '3', '5', '7']	['2', '3', '5', '7']	[1459, 1436, 1346, 1108]	1337.25	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1459
21	2	21		['4', '6', '7', 'T', 'R']	['4', '6', '7', 'T', 'R']	[1389, 1332, 1395, 1060, 106	1248.4	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1389
22	2	21		['2', '3', '5', '8', '9']	['2', '3', '5', '8', '9']	[2318, 2295, 2214, 2274, 227	2274.6	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	2318
23	2	21		['3', '5', '7', '8', '9']	['3', '5', '7', '8', '9']	[1284, 1364, 1537, 1524, 136	1414.4	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1284
24	2	21		['2', '9', 'A', 'R']	['2', '9', 'A', 'R']	[2339, 2113, 2220, 2004]	2169	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	2339
25	2	21		['2', '4', '6', '8']	['2', '4', '6', '8']	[1242, 1334, 1352, 1196]	1281	TRUE	4	4	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 08:45:40	1242
26	3	25		['2', '5', '6', 'M', 'S'	['2', '5', '6', 'M', 'S']	[2066, 2225, 2024, 2263, 240	2197.2	TRUE	5	5	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	2066
27	3	25		['2', '4', '6', '8']	['2', '4', '6', '8']	[1276, 1239, 1339, 1542]	1349	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	1276
28	3	25		['2', '3', '6', '8']	['2', '3', '6', '8']	[1933, 1983, 2040, 2014]	1992.5	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	1933
29	3	25		['2', '3', '5', '8', '9']	['2', '3', '5', '8', '9']	[2154, 2257, 2392, 2091, 228	2234.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	2154
30	3	25		['2', '3', '5', '8', '9']	['2', '3', '5', '8', '9']	[2023, 1790, 1935, 1822, 194	1902.8	TRUE	5	5	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	2023
31	3	25		['3', '4', '6', '7', '9']	['3', '4', '6', '7', '9']	[2846, 2532, 2643, 2837, 249	2670.2	TRUE	5	5	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	2846
32	3	25		['4', '7', 'T', 'R']	['4', '7', 'T', 'R']	[1103, 1312, 1235, 1019]	1167.25	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	1103
33	3	25		['4', '7', 'T', 'R']	['4', '7', 'T', 'R']	[1239, 1204, 1176, 1018]	1159.25	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	1239
34	3	25		['2', '4', '6', '8']	['2', '4', '6', '8']	[1590, 1767, 1656, 1737]	1687.5	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	1590
35	3	25		['2', '3', '5', '7']	['2', '3', '5', '7']	[2523, 2429, 2314, 2419]	2421.25	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	2523
36	3	25		['2', '4', '6', '8']	['2', '4', '6', '8']	[1980, 1901, 2021, 1860]	1940.5	TRUE	4	4	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	1980
37	3	25		['4', '6', '7', 'T', 'R']	['4', '6', '7', 'T', 'R']	[4161, 4343, 4140, 4347, 439	4276.6	TRUE	5	5	29-04-25 13:25:4	29-04-25 09:25:4	29-04-25 09:49:40	4161
38	4	29		['2', '3', '5', '8', '9']	['2', '3', '5', '8', '9']	[2971, 2959, 2983, 2626, 279	2866.2	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 09:05:40	2971
39	4	29		['2', '5', '6', 'M', 'S'	['2', '5', '6', 'M', 'S']	[1310, 1226, 1300, 1450, 155	1367.6	TRUE	5	5	29-04-25 13:25:4	29-04-25 08:25:4	29-04-25 09:05:40	1310

Test Results cont.

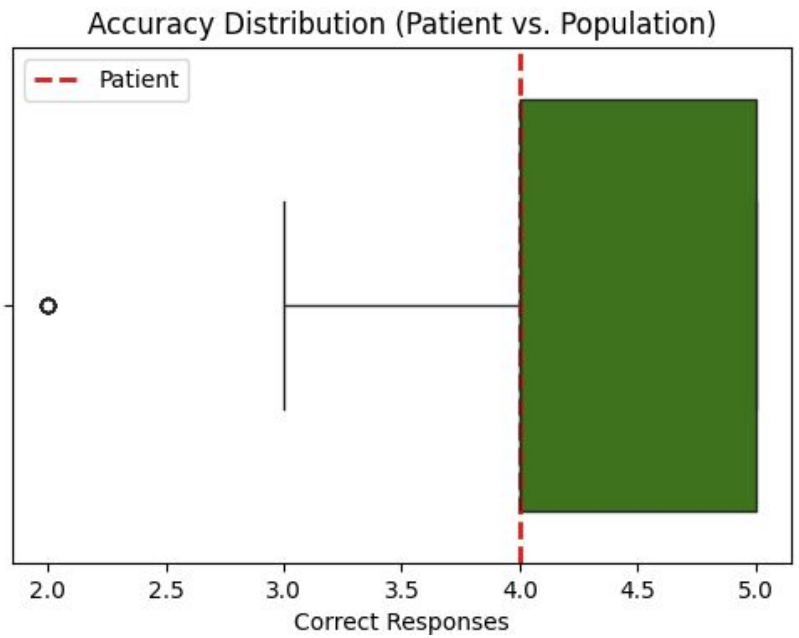
Aggregated Statistics cont.

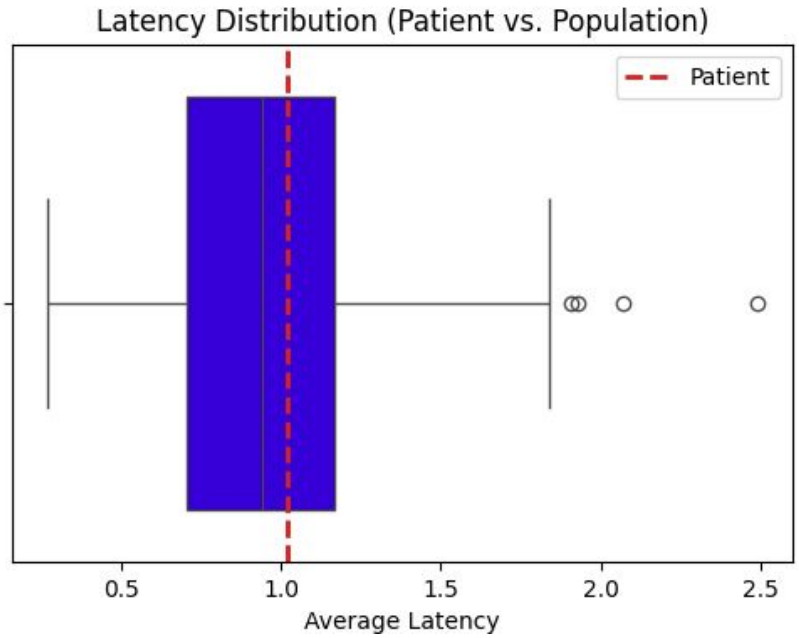
	A	B	C	D	E	F	G	H	I	J	K	L
1	Test ID	Accuracy Rate	Avg Count Correct	Mean Latency	Latency Std Dev	Median First Latency	Fastest First Latency	Slowest First Latency	Latency-Accuracy Corr	Proportion Partial Success	Accuracy by Span	Latency by Span
2	1	1	4.58	2330.51	1064.468	2157.5	1163	4271		0	{4: 1.0, 5: 1.0}	{4: 2098.95, 5: 2495.91}
3	2	1	4.42	1807.66	644.754	1492	1091	3252		0	{4: 1.0, 5: 1.0}	{4: 1737.93, 5: 1905.28}
4	3	1	4.42	2083.24	839.656	2001.5	1103	4161		0	{4: 1.0, 5: 1.0}	{4: 1673.89, 5: 2656.32}
5	4	1	4.58	2364.7	1055.712	2342	1253	4242		0	{4: 1.0, 5: 1.0}	{4: 2786.85, 5: 2063.17}
6	5	0.917	4.5	2634.34	705.148	2722.5	1608	3900	-0.298	0.08	{4: 1.0, 5: 0.88}	{4: 2627.56, 5: 2637.72}
7	6	1	4.42	2786.04	1006.187	2965.5	1084	4387		0	{4: 1.0, 5: 1.0}	{4: 2919.61, 5: 2599.04}
8	7	0.833	4.08	3035.63	703.574	2893	2167	4869	0.188	0	{4: 0.71, 5: 1.0}	{4: 3096.25, 5: 2950.76}
9	8	0.75	4.08	2635.65	798.41	2528	1408	3901	-0.27	0.25	{4: 1.0, 5: 0.57}	{4: 2130.2, 5: 2996.69}
10	9	0.833	4.08	2975.15	621.627	2788	1828	4143	0.272	0.08	{4: 0.86, 5: 0.8}	{4: 3191.46, 5: 2672.32}
11	10	0.667	3.92	3474.62	1075.617	3750.5	1645	4984	0.547	0.08	{4: 0.4, 5: 0.86}	{4: 3612.7, 5: 3376.0}
12	11	0.583	3.83	3274.54	701.049	3464	1434	4062	0.191	0.17	{4: 0.25, 5: 0.75}	{4: 2893.06, 5: 3465.28}
13	12	0.75	4.17	3272.65	607.345	3412	2365	4179	-0.237	0.17	{4: 0.75, 5: 0.75}	{4: 3296.0, 5: 3260.98}
14	13	0.5	3.42	3635.41	699.323	3549	2615	4783	0.177	0.17	{4: 0.43, 5: 0.6}	{4: 3820.64, 5: 3376.08}
15	14	0.333	3.33	3492.33	816.6	3531.5	2311	5150	-0.209	0.42	{4: 0.25, 5: 0.38}	{4: 3639.19, 5: 3418.9}
16	15	0.5	3.5	4078.4	756.474	3821	2930	5124	-0.141	0.25	{4: 0.5, 5: 0.5}	{4: 4214.62, 5: 3942.17}
17	16	0.167	2.58	3673.46	754.441	3698	2546	4949	0.005	0.17	{4: 0.11, 5: 0.33}	{4: 3649.97, 5: 3743.93}
18	17	0.333	3.42	4207.52	556.746	4338.5	3071	4921	-0.106	0.58	{4: 0.67, 5: 0.22}	{4: 3802.33, 5: 4342.58}
19	18	0.5	3.5	3903.75	878.977	3854	2436	4941	-0.041	0.25	{4: 0.5, 5: 0.5}	{4: 3279.21, 5: 4528.3}
20	19	0.083	2.75	4122.6	707.226	4001.5	3005	5155	-0.19	0.5	{4: 0.0, 5: 0.14}	{4: 3902.25, 5: 4280.0}
21	20	0.333	3.25	4071.9	576.601	3869.5	3365	5140	-0.119	0.25	{4: 0.0, 5: 0.57}	{4: 4127.75, 5: 4032.0}
22	31	0.688	3.75	980.42	173.281	1384	835	3737	-0.067	0.19	{4: 0.88, 5: 0.5}	{4: 998.06, 5: 962.78}
23	Overall	0.655714286	3.837142857	3087.64381	749.6769524	3074.452381	1964.904762	4488.142857	-0.018625	0.171904762		

More to be shown during live demo...

Plots

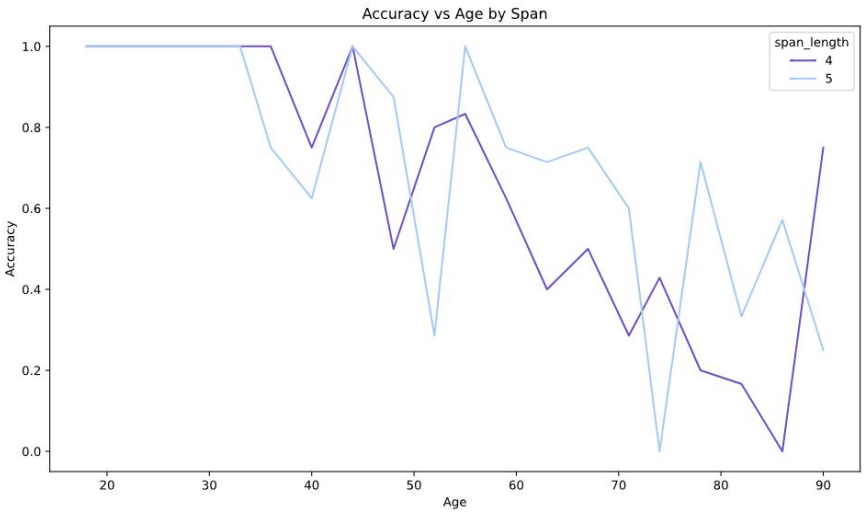
Individual plots are automatically generated by the server after each test to visualize key performance metrics. Using Seaborn and Matplotlib, the server creates graphs showing response times, accuracy rates, and response patterns. These plots help the test giver quickly interpret the results and are not visible to the test taker.

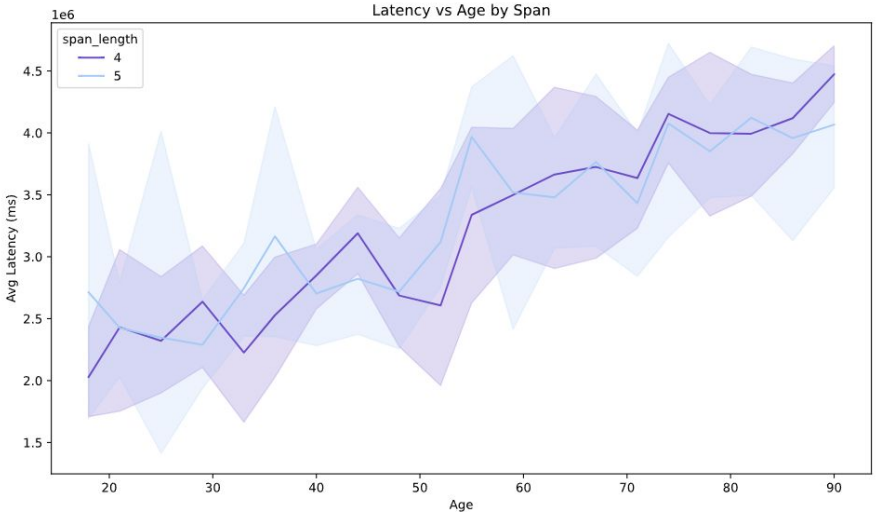




Plots cont.

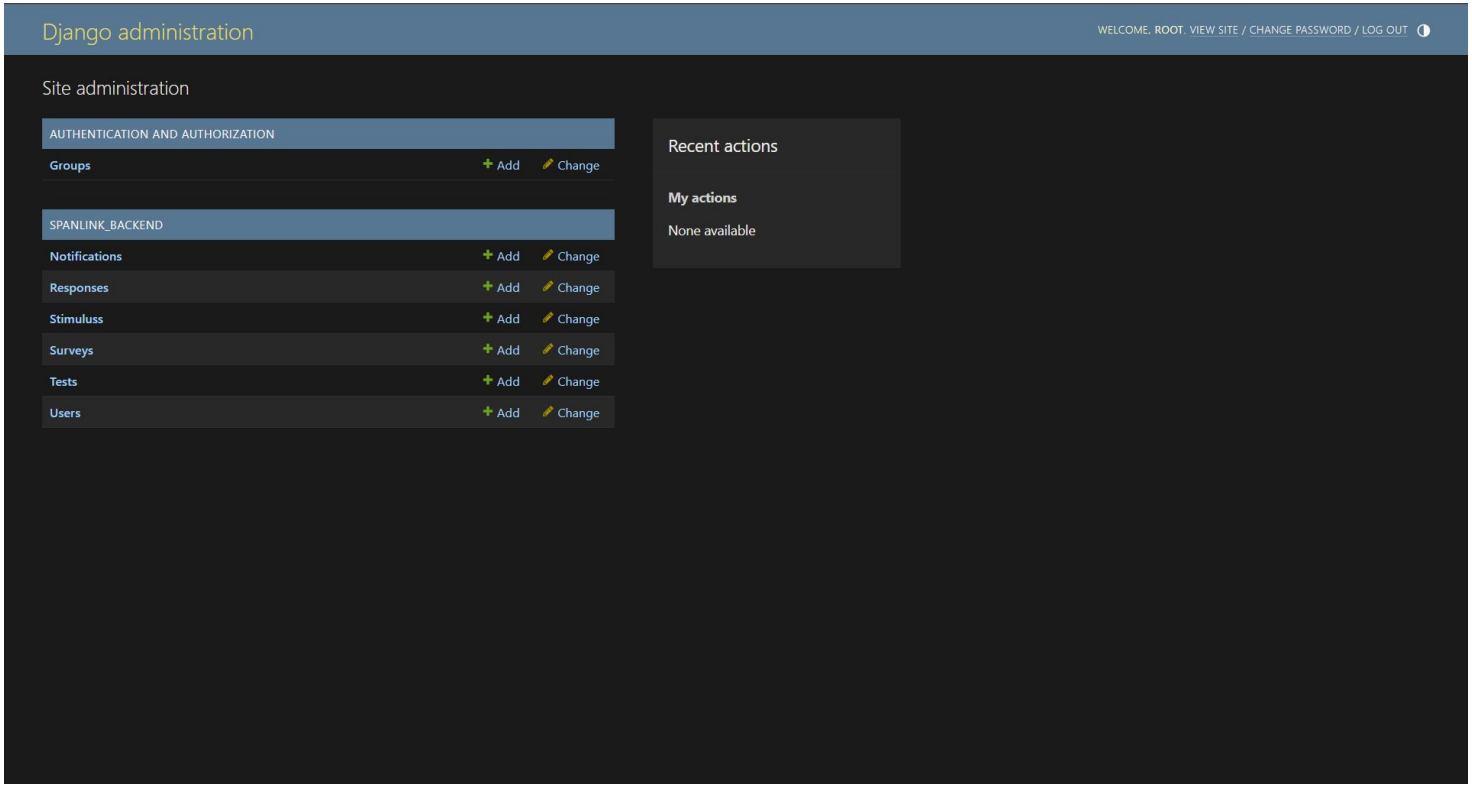
Aggregated plots are generated, such as average latency and accuracy by age group and span length, latency differences between correct and incorrect responses, and trends in performance by age. These visualizations highlight broader patterns in user behavior and help interpret results across the entire dataset.





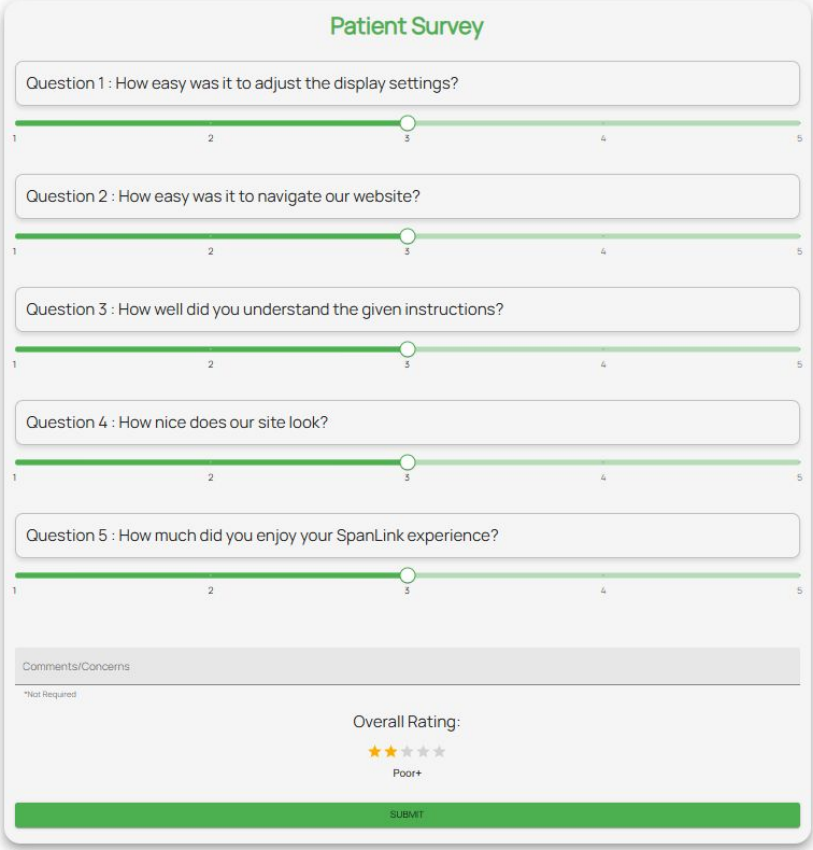
Administrative Capabilities

8. Administrators will be able to login to an administration portal and view statistics and manage the database of test givers.



Survey

When the user has completed their prescribed test, they will be presented with a prompt to take an optional survey. If the user decides to opt in, they will be taken to this page where they can rate their experience and leave comments. When the user submits their survey, a prompt will appear, lock the screen, and notify the user that their survey has been submitted successfully, and that they are free to close the window.



03

Live Demo

http://localhost:3000/

Questions?

THANK YOU

From the
SpanLink Team